

A systematic assessment of the performance gains of baseline and fine-tuned protein language models vs. classic features for the downstream task of epitope prediction

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Abstract

Linear B-cell epitopes (LBCEs) are short, continuous amino acid sequences on an antigen's surface that are recognised and bound by specific antibodies or B-cell receptors. Identifying epitopes plays a key part in the development of therapeutics and diagnostics, however experimental methods for their identification are cost and resource intensive. Computational methods that predict LBCE sequences can screen high volumes of protein sequences at low cost. Traditionally, machine learning models for LBCE prediction used inputs derived through engineered features from amino acid physiochemical and statistical properties. More recently, deep learning architectures, trained using masked language modelling have been leveraged to generate protein language models (pLMs). These models provide rich semantic representations that can capture the linear sequences of a protein with general structural, functional and evolutionary context and can be adapted to epitope prediction tasks using transfer learning. Despite the intuitive appeal of pLMs, the quantitative gains in predictive performance from adopting these models have not yet been characterised in the literature. This project will investigate this by performing a systematic comparison of the predictive performance of models derived from pLMs (both as-standard and fine-tuned to the task of LBCE prediction) and equivalent models built based on the classical engineered features. The hypothesised outcome of this project is that fine-tuning can be used to apply computationally efficient transfer learning to large-scale pLMs for epitope prediction. These predictions are likely to surpass the performance of features generated through both fully fine-tuned lower parameter pLMs and classical engineered feature methods.

Ethics statement: This project fits within the scope of the blanket ethics application, as reviewed by my supervisor Felipe Campelo. I have completed the ethics test on Blackboard. My score is 12/12.

1 Project Plan

Linear B-cell epitopes (LBCs) are short, continuous amino acid sequences on an antigen’s surface that are recognised and bound by specific antibodies or B-cell receptors [1, 2, 3]. The epitope/B-cell receptor interaction stimulates an immune response [4]. Identifying epitopes is key part in the development of new vaccines, therapeutic antibodies and immunodiagnostics [3, 5]. B-cell epitopes are broadly categorised into linear and conformational, where linear B-cell epitopes (LBCs) are short sequences of amino acids that are adjacent in an antigen’s primary sequence, while conformational B-cell epitopes are formed from non-adjacent amino acids brought together by the folding (tertiary structure) of an antigenic protein [6, 7]. Experimental epitope identification includes techniques like X-ray crystallography, nuclear magnetic resonance spectroscopy and peptide microarrays [5, 8]. These methods underpin much of the current scientific understanding of protein biochemistry and have generated vast databases of protein data [9, 10]. From this data, computational methods that aim to predict epitopes have been developed, whilst often circumventing the cost and resource implications of traditional experimental techniques [3, 6, 7, 11].

Early computational methods for B-cell epitope identification focused on comparing sequence similarity to known epitopes, known as multiple sequence alignment (MSA) [8, 9]. This method is predicated on the likelihood of sequence conservation through evolution, therefore informing an increased chance of functional similarity between previously identified and potential epitopes [8, 9]. Consequently, MSA is limited to the comparison of functionally annotated sequences [8] and may struggle with novel sequences such as those from understudied or emerging pathogens.

Another early method, used numeric scales of amino acid propensities, like degree of hydrophilicity, to describe potential epitope sequences. This method converts protein primary sequences into numerical values representing residue propensities, and applies moving averages and sliding windows for different polypeptide lengths. Resulting peaks are then assigned as potential epitopes. A comprehensive assessment of the method applied 484 different propensity scales to the primary structures of 50 proteins with confirmed epitope sequences. They found that propensity peak sequences and epitope sequences were not significantly correlated. Since epitope-antibody interactions involve complex physiochemical contexts, the poor predictive performance from using a single propensity measure is unsurprising. Subsequently, feature engineering tools like Protr and Pfeature [8] which integrate evolutionary data and experimentally informed residue-level annotations, produce detailed and comprehensive numeric descriptions of amino acids in sequence. These features have been used as inputs to machine learning models such as inputs to predict pattern recognition receptors [8].

More recently, the same deep learning approaches used in large language models (LLMs) have been leveraged to develop protein language models (pLMs) [7, 9, 12]. Following downstream transfer learning, pLMs can be used for a wide diversity of predictive tasks, including epitope prediction. These models are pre-trained through systematic masking of amino acid tokens as the model learns to predict the masked token in context of the input sequence [9, 12]. Model weights are adjusted using backpropagation to minimise a cost function associated with the probability of the predicted amino acid and the true masked amino acid [12, 13]. The pre-trained pLM generates residue-level embeddings that represent a rich semantic meaning of a protein with long-range context [14, 15]. These embeddings capture both the linear sequence of a protein but also a representation of its general structural, functional and evolutionary context [12, 13, 16]. Drawing conclusions from these representations must consider the quality of the input data and architecture of the model, in particular, the architecture must balance the computational cost and embedding expressiveness which both increase with model size [9, 13].

pLMs can be adapted for epitope prediction through supervised transfer learning [3, 11]. The simplest method of transfer learning adds additional machine learning classifiers or heads to the pLM, trained using a dataset of known epitope and non-epitope sequences [11]. To enhance predictive performance, the underlying pLM can be fine-tuned [9, 11]. Full fine-tuning updates the pLM weights to the downstream predictive task, however as large protein language models contain billions of parameters, full fine-tuning becomes increasingly computationally prohibitive [9, 10]. Parameter efficient fine tuning (PEFT) can overcome this limitation by adding smaller matrices into the model that apply a change to the pre-trained weights, resulting in fine-tuned model at a much lower training cost [9, 17]. One such method is low-rank adaptation (LoRA), which uses a low-rank decomposition of the original weights to add two additional matrices to the self-attention modules within each transformer block [17]. Previous work has shown that LoRA provides competitive prediction performance to full fine-tuning for protein prediction tasks whilst training only 0.25% of the model parameters [9].

Given that epitope sequences are often highly conserved between species due to genetic similarity from common ancestry, particular care must be taken when segregating epitope sequences for model development and testing to prevent data leakage and artificial predictions [4, 5, 11]. Previous work has successfully used phylogeny-informed fine tuning of a protein language model from data-rich pathogens to generate accurate predictions for data-scarce pathogens [11]. Although PEFT has been applied to the transfer learning of pLMs to epitope [18] and antigen specificity [10] predictions, evidence of its application to phylogeny-informed LBCE prediction is unknown.

Using the pLMs referenced by Schmirler et al., [9] this project will complete a comparative assessment of pLMs deployed for transfer learning of LBCE prediction tasks, incorporating phylogeny-aware data splits during training and testing. The 3 different pLMs used within [9] are ESM2 [12], Ankh [19] and ProtT5/ProstT5 [20]. Each pLM type consists of a family of models of varying depth and parameter size. Architectures also vary between models, for example ESM2 models are encoder only models that produce embeddings of input sequences whereas Ankh and ProstT5/ProstT5 models are encoder-decoder models that additionally can generate forward sequence predictions. Furthermore, ProstT5 can also predict protein structures [9].

pLM transfer learning will use 3 different approaches. For each approach, initial transfer learning will be applied using higher-level data from a data rich pathogen followed by model fitting and testing using lower-level data. To assume a baseline performance, the standard pre-trained pLM with no fine-tuning will be applied to epitope prediction using a classification head trained on higher-level pathogen data. For models with fewer parameters, full fine-tuning will be performed using a classification head with the higher-level data, with model fitting and testing completed using the lower-level data. For all model sizes, PEFT using LoRA will be performed. Performance will be compared across fine tuning methods and baseline models, with additional data on computation cost obtained. Finally, for comparison to classic domain-specific feature generation methods, amino acid features will be generated using Protr or Pfeature and used as input to an epitope classification head.

The hypothesised outcome is that PEFT can be used to apply computationally efficient transfer learning to multibillion parameter pLMs for epitope prediction. These predictions are likely to surpass the performance of features generated through both fully fine-tuned lower parameter pLMs and classic domain specific feature generation methods.

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A Project Timeline

A timeline for the project activities is shown in Figure 1.

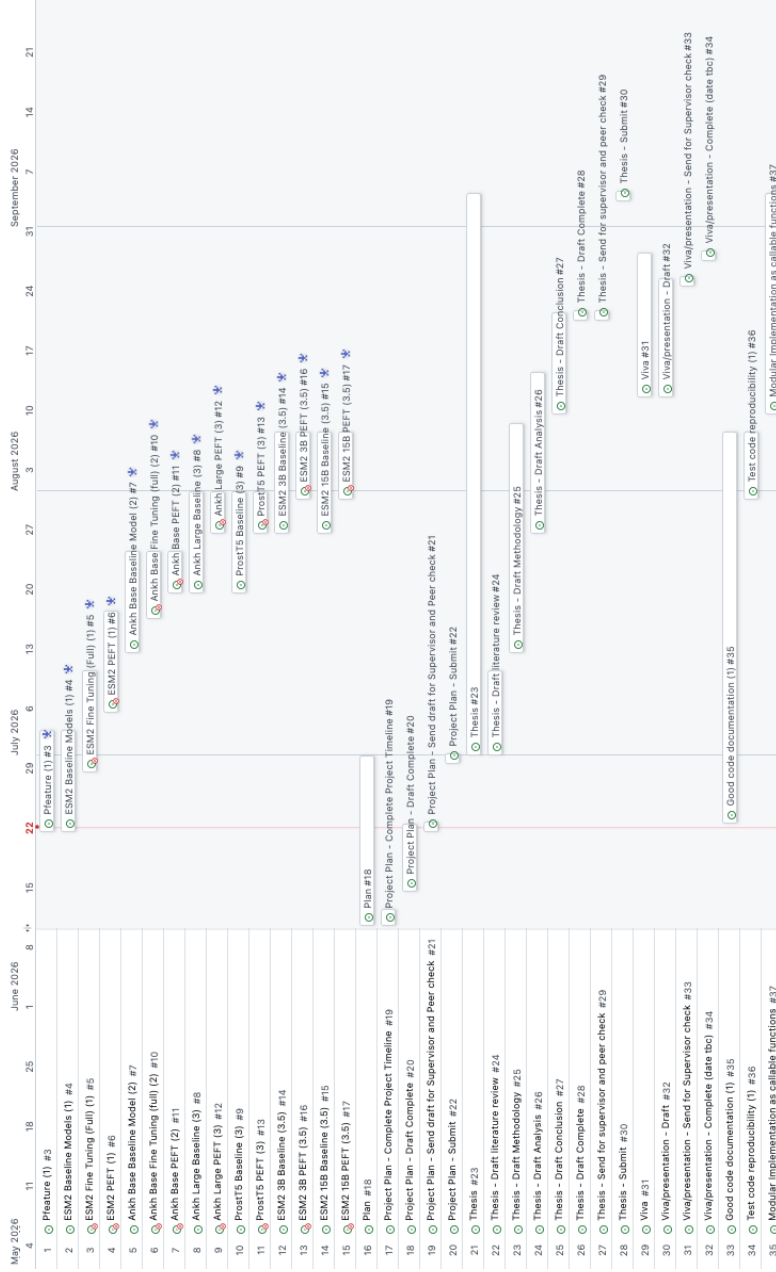


Figure 1: Project roadmap showing key milestones and deliverables.

B Risk Assessment

Table 1 shows the risk assessment for this project.

Table 1: Risks and Mitigations

Risk	Likelihood	Severity	Mitigation
Computer stolen or broken	Medium	High	Backup all work using the cloud.
Insufficient Compute Resource to fine tune / run models	High	High	Use cloud based GPUs for model training and testing. AWS resources were requested. Failing that, use supervisor's Isambard-AI budget.
Data leakage between train / validation / test splits	High	High	Ensure sequences are adequately split according to predefined sequence similarity criteria
Insufficient time for supervisor review of Thesis	Medium	Medium	Include thesis review date in project plan
Model underperformance	Medium	Low	Additional time included in plan for model development, results documented and compared as empirical assessment for project. Focus study on performance quantification and hypothesis testing rather than on raw performance maximisation.